Practice Exam 2 (F25)

Started: Nov 2 at 9:59am

Quiz Instructions

Question 1 1 pts

(x,y) = (0,0) is located at the [Select] [Select] of the LCD screen

(x,y) = (0,127) is located at the [Select] [Select] of the LCD screen

(x,y) = (127,0) is located at the [Select] [Select] of the LCD screen

(x,y) = (127,127) is located at the [Select] [Select] of the LCD screen

Question 2 1 pts

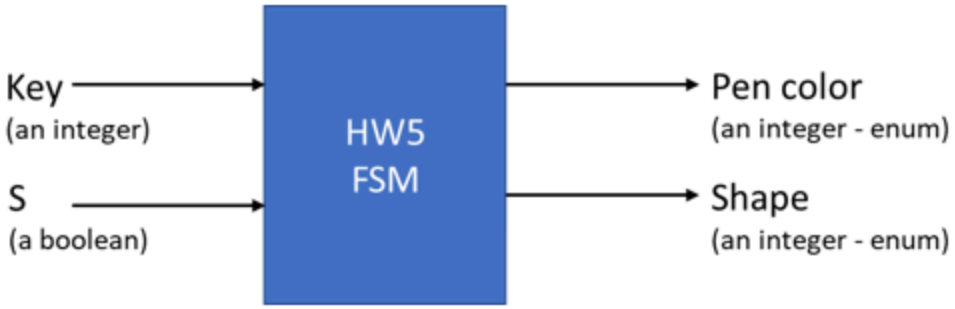
Which of the following will draw a horizontal line on the right side of the screen, one quarter of the way up from the bottom?



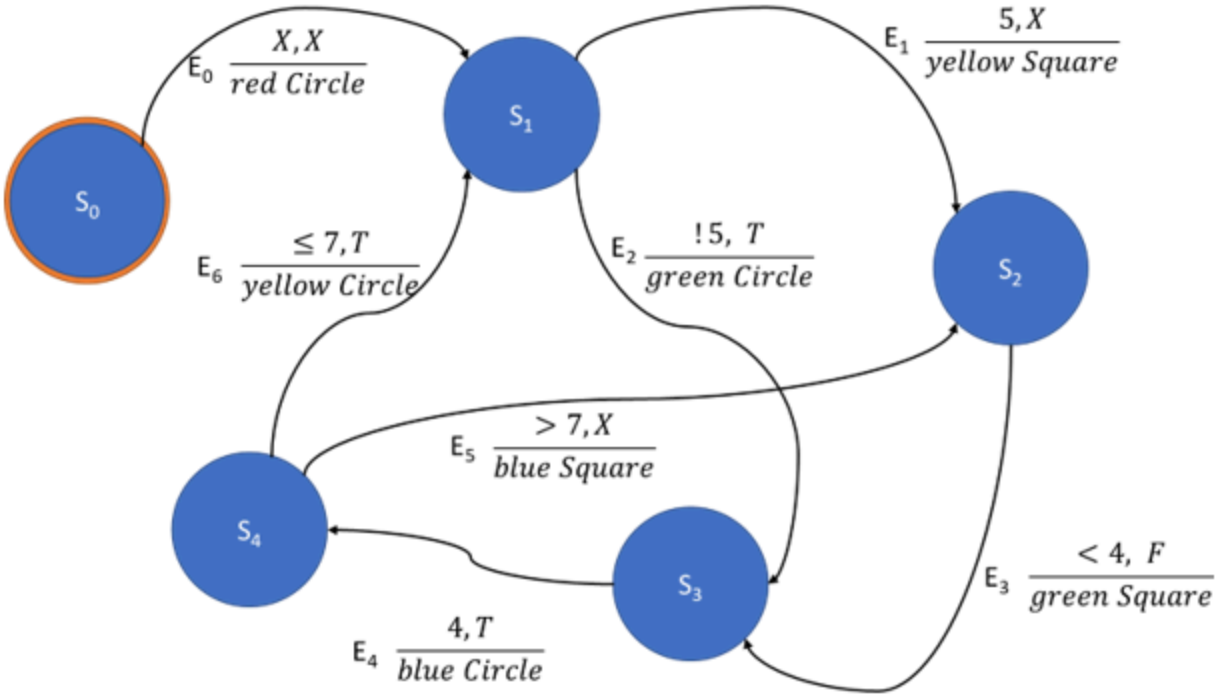
- Graphics_drawLineH(&g_sContext, 96, 64, 127);
- Graphics_drawLineH(&g_sContext, 32, 64, 127);
- Graphics_drawLineH(&g_sContext, 64, 127, 32);
- Graphics_drawLineH(&g_sContext, 64, 127, 96);

Consider the below figure for the following questions:

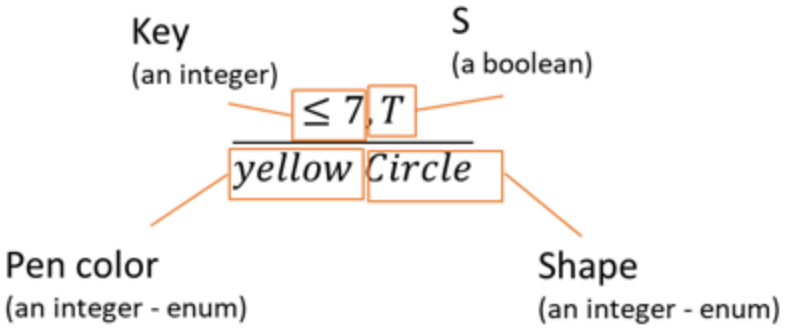
Block diagram:



FSM diagram:



Transition (edge) format:



Question 3 1 pts

What is the number of inputs of this FSM?

Question 4 1 pts

For the same FSM as above, what is the number of outputs?

Question 5 1 pts

For the same FSM as above, what is the number of states?

Question 6 1 pts

For the same FSM as above, what is the number of transitions?

Question 7 1 pts

In the same FSM, starting from the initial state, assume the following series of inputs are given to the FSM.

key = 5 and S = true

key = 5 and S = false

key = 2 and S = false

What is the final output at this point?

Your answer should be in the form (color) (shape)

Example: red square

Question 8 1 pts

SPI is [Select] , I2C is [Select] , and UART is [Select] .

Question 9 1 pts

SPI is [Select] , I2C is [Select] , and UART is [Select] .

Question 10 1 pts

SPI is [Select] , I2C is [Select] , and UART is [Select] .

Question 11 1 pts

For a UART module with (8N1) format (which means 8 data bits, No parity and 1 stop bit), and Baudrate of 1000 Baud-per-second. Assume the channel is used at its full capacity. choose all the true statements:

1. The bitrate is 1000 bit/sec. [Select]
2. The data rate is 1000 bit/sec. [Select]
3. The data rate is 800 bit/sec. [Select]
4. The bitrate is 2000 bit/sec. [Select]

If the the BRCLK's frequency (BRCLK is the input clock of the UART) is 500KHz and Baudrate is 115200, determine the following configuration parameter values.

Question 12 1 pts

What is the prescalar of the above configuration?

Question 13 1 pts

What is the firstMod of the above configuration?

Question 14 1 pts

What is the secondMod of the above configuration?

Answer in the form 0x##

Question 15 1 pts

What is the oversampling value of the above configuration?

Question 16 1 pts

In terms of seconds, how long is 2 million cycles of a processor with a clock frequency of 1MHz?

Question 17 1 pts

For a processor with a clock frequency of 1 GHz, how many cycles exist in 3 seconds?

Question 18 1 pts

If the system clock frequency feeding a timer is 4MHz and the Prescaler value is 16, what is the counter clock period (in microseconds)?

Question 19 1 pts

Assume the system clock is clocked at 3000000 Hz.

Assume the following lines are used to initialize a Timer32.

Timer32_initModule(TIMER32_0_BASE,
UNKOWN,
TIMER32_32BIT,
TIMER32_PERIODIC_MODE);

Assume the following lines are used to start the timer in the one-shot mode.

Timer32_setCount(TIMER32_0_BASE, 12000);
Timer32_startTimer(TIMER32_0_BASE, true);

If we wish for the timer to expire after *about* 1s, what prescalar value should we use in the UNKNOWN line to satisfy this (just a number).

Question 20 1 pts

For Timer32, the largest possible Prescaler value is 256 and the smallest Prescaler value is 1. What is the longest interval **(in seconds)** Timer32 can measure if the system clock frequency (the one that feeds the timer) is 48MHz? Give your answer as a rounded integer +/1 error is acceptable.

Question 21 1 pts

What is the binary representation of the ASCII character 'Q'?

Your answer should be 8 bits long, include leading 0s.

Question 22 1 pts

Using the diagram below, determine the values of the following variables at the moment of Event 2 within the function SWTimer_elapsedCycles(SWTimer* timer_p).

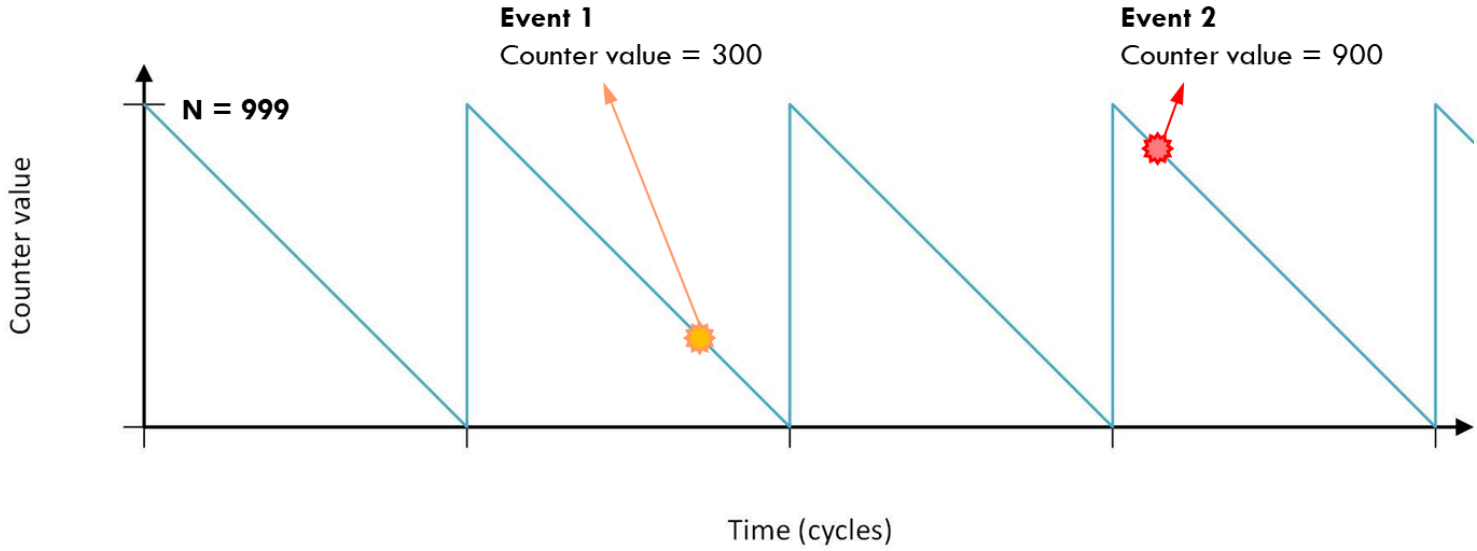
hwTimerRollover =

timer_p->startRollovers =

currentCounter =

timer_p->startCounter =

SWTimer_elapsedCycles =



Question 23 1 pts

Consider the following snippet of code:

typedef enum { Apple, Banana, Carrot } food_t;

What number is Carrot assigned in this struct? [t1]

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